ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** silver	Silver lies below hydrogen.
2	**(b)** They have high melting points	Strong forces between ions.
3	**(b)** keeps air and water away	Barrier effect.
4	**(b)** sodium	Only highly reactive metals react with cold water.
5	**(b)** +2	It loses two electrons.
6	**(b)** is more reactive than iron	Zinc corrodes first.
7	**(b)** $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$	Only a more reactive metal displaces a less reactive.
8	**(b)** X^-	It gains one electron.
9	**(b)** anodising	Anodising.
10	**(b)** Mg > Zn > Fe > Cu	Standard reactivity series.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Bubbles of hydrogen stick to its surface (1); They make it lighter than water (1).	1+1
12	Ions are free in solution (1); Fixed in the solid lattice (1).	1+1
13	Water is polar and separates ions (1); Kerosene is non-polar (1).	1+1
14	Copper is below hydrogen in the series (1); Zinc is above hydrogen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$ (1); A thin layer of Al_2O_3 forms on the surface (1); It prevents further reaction (anodising thickens it) (1).	1+1+1
16	Crushing and grinding (1); Removal of gangue (1); Roasting converts ZnS to ZnO (1).	1+1+1
17	Zinc layer isolates iron from air (1); Zinc is more reactive and gets oxidised first (1); So iron is protected even if scratched (1).	1+1+1
	OR: A > C > B (1); A displaces hydrogen from acid and reacts with cold water (1); B does not react with acids, so it is least reactive (1).	1+1+1
18	Metals from reduction contain impurities (1); Electrolytic refining: impure Cu anode and pure Cu cathode (1); Pure copper deposits on the cathode (1).	1+1+1
	OR: Position in the reactivity series (1); Na, K are at the top (1); Cu is below hydrogen (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Metal oxides are basic (1); Hydrogen is displaced by reactive metals (1).	1+1+1+1+1
	OR: Bridges: painting, galvanisation (1); Aluminium: anodising (1); Ships: sacrificial zinc/magnesium plates (1); Zn, Mg are more reactive than iron (1); They corrode instead of iron (1).	1+1+1+1+1
20	Hg: heating cinnabar, as unreactive (1); $2\,HgS + 3\,O_2 \longrightarrow 2\,HgO + 2\,SO_2$; $2\,HgO \longrightarrow 2\,Hg + O_2$ (1); Zn: calcination and reduction with carbon (1); $ZnO + C \longrightarrow Zn + CO$ (1); Na: electrolysis of molten NaCl (1).	1+1+1+1+1
	OR: Zn displaces Fe and Cu (1); Fe displaces Cu but not Zn (1); Cu displaces none (1); Order Zn > Fe > Cu (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $MgCl_2$	Mg^{2+} and Cl^-.
2	**(b)** heating in air	Heating in air promotes oxidation.
3	**(b)** hydrogen	HNO_3 oxidises the hydrogen formed.
4	**(c)** Na_2O	Na^+ and O^{2-}.
5	**(c)** a thin protective layer forms on its surface	A passive protective film forms.
6	**(d)** mercury	Mercury is unreactive.
7	**(d)** like charges come close on shifting layers	Repulsion causes breaking.
8	**(d)** a thin strip of pure copper	Cu^{2+} deposits on it.
9	**(d)** sulphides and carbonates	Zn, Fe, Pb ores.
10	**(a)** NaCl	Ionic compounds.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Roasting: sulphide ore in excess air (1); Calcination: carbonate ore in limited air (1).	1+1
12	Iron displaces copper (1); Copper deposits on the nail (1).	1+1
13	Sodium is more reactive than carbon (1); Carbon cannot reduce its oxide; electrolysis is used (1).	1+1
14	Highly reactive, forms stable compounds (1); Only electrolytic reduction can reduce Na^+ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	X: 2,8,2 loses 2 electrons; Y: 2,8,7 gains 1 (1); XY_2 (i.e. $MgCl_2$) (1); Ionic bond (1).	1+1+1
16	Alloys are harder and stronger (1); Adding carbon makes iron hard and brittle (1); Ni and Cr make stainless steel which resists rusting (1).	1+1+1
17	React with acids and bases (1); Al_2O_3, ZnO (1); $Al_2O_3 + 6\,HCl \longrightarrow 2\,AlCl_3 + 3\,H_2O$ (1).	1+1+1
	OR: Mg: vigorous bubbles (1); Zn: moderate (1); Cu: no reaction as it is less reactive than hydrogen (1).	1+1+1
18	$3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (1); $Ca + 2\,H_2O \longrightarrow Ca(OH)_2 + H_2$ (1); Hydrogen is evolved in both (1).	1+1+1
	OR: Copper: basic copper carbonate (1); Silver: silver sulphide (1); Due to reaction with air gases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	MO is amphoteric (e.g. ZnO) (1); $ZnO + 2\,HCl \longrightarrow ZnCl_2 + H_2O$ (1); $ZnO + 2\,NaOH \longrightarrow Na_2ZnO_2 + H_2O$ (1); M can be zinc (1); Aluminium is another such metal (1).	1+1+1+1+1
	OR: $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$ (1); Al is more reactive than Fe (1); Aluminium reduces Fe_2O_3 (1); Highly exothermic; iron is molten (1); Used for welding rails (1).	1+1+1+1+1
20	Na: violent and exothermic (1); Ca: less violent, floats (1); Mg: reacts with hot water (1); Al, Fe: only with steam (1); Difference arises from reactivity (1).	1+1+1+1+1
	OR: Ionic: high melting point; NaCl (1); Covalent: low; CH_4 (1); Ionic conduct in solution (1); Covalent do not (1); Ionic soluble in water; covalent in organic solvents (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** silver and gold	Less reactive metals do not dissolve.
2	**(b)** aluminium	Al is more reactive than Fe.
3	**(a)** carbon	Non-metals share electrons.
4	**(c)** liquid and expands uniformly	Property of mercury.
5	**(a)** sodium	$Na^+ + e^- \longrightarrow Na$.
6	**(c)** 3	Al (2,8,3) loses 3.
7	**(c)** solder	Low melting point.
8	**(b)** aluminium	Al_2O_3 layer.
9	**(a)** $AlCl_3$	Al^{3+} and Cl^-.
10	**(b)** sodium chloride	It is a compound.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Welding railway tracks (1); $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$ (1).	1+1
12	Black coating of copper(II) oxide (1); $2\,Cu + O_2 \longrightarrow 2\,CuO$ (1).	1+1
13	$ZnO + 2\,NaOH \longrightarrow Na_2ZnO_2 + H_2O$ (2).	2
14	24-carat gold is pure gold (1); 22-carat is alloyed with copper/silver to be harder (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Aqua regia is 3 parts HCl to 1 part HNO_3 (1); HNO_3 oxidises gold; chloride forms soluble complex (1); Gold dissolves slowly (1).	1+1+1
16	Sodium chloride (1); Electron transferred from Na to Cl (1); Electrostatic force makes an ionic bond (1).	1+1+1
17	Flux combines with gangue (1); Slag is formed (1); Slag is removed (1).	1+1+1
	OR: $2\,Cu_2S + 3\,O_2 \longrightarrow 2\,Cu_2O + 2\,SO_2$ (1); $2\,Cu_2O + Cu_2S \longrightarrow 6\,Cu + SO_2$ (1); Roasting then auto-reduction (1).	1+1+1
18	Metals have free electrons (1); Graphite layers have delocalised electrons (1); Diamond has none (1).	1+1+1
	OR: (i) Iron displaces copper (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1); (ii) No reaction, since zinc is more reactive (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mg loses 2, O gains 2 (1); Na loses 1; O gains 2, so Na_2O (1); High melting points (1); Brittle (1); Conduct in molten state (1).	1+1+1+1+1
	OR: Alloys are harder and more resistant (1); Gold is soft (1); 22-carat has Ag/Cu added (1); Brass (1); Bronze (1).	1+1+1+1+1
20	Anode: impure copper (1); Cathode: pure copper (1); Electrolyte acidified $CuSO_4$ (1); Copper dissolves from anode and deposits on cathode (1); Impurities settle as anode mud (1).	1+1+1+1+1
	OR: $Al_2O_3 + 2\,NaOH \longrightarrow 2\,NaAlO_2 + H_2O$ (1); $Al_2O_3 + 6\,HCl \longrightarrow 2\,AlCl_3 + 3\,H_2O$ (1); Zn reacts with acid and base (1); $Zn + 2\,NaOH \longrightarrow Na_2ZnO_2 + H_2$ (1); Oxide layer protects metal (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.